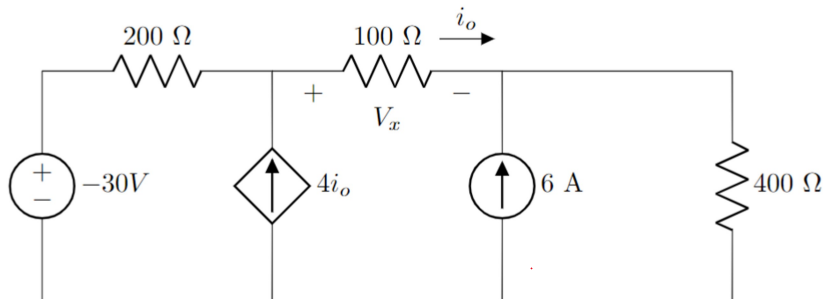


Semester Test 2 Memo RV2

Friday, 27 May 2022 20:17

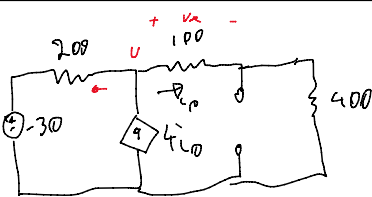
Questions 1 to 2 [4]

Use superposition to determine the contribution of the independent voltage source to the voltage V_x exclusively and the exclusive contribution of the independent current source to the voltage V_x respectively.



01 Voltage source contribution V_{xV} . [2] **30V**

02 Current source contribution V_{xI} . [2] **160V**



KCL

$$\frac{v - (-30)}{200} + \frac{v - 0}{600} = 4i_o$$

$$i_o = \frac{v - 0}{500}$$

$$\Rightarrow \frac{v - (-30)}{200} + \frac{v}{600} = \frac{4v}{500}$$

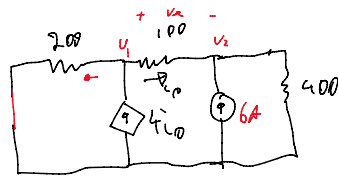
$$\times 1500 \Rightarrow 5v + 150 + 2v = 8v$$

$$\Rightarrow 150 = 8v - 7v = v$$

$$\Rightarrow i_o = \frac{150}{500} = 0.3 \text{ A}$$

$$V_x = i_o \times 100\Omega = 0.3 \times 100 = 30 \text{ V}$$

$$V_x = 30 \text{ V}$$



$$i_o = \frac{V_1 - V_2}{100} \quad V_x = V_1 - V_2 \quad \text{or} \quad i_o \times 100$$

$$\textcircled{1} \quad \frac{V_1}{200} - 4i_o + \frac{V_1 - V_2}{100} = 0 \quad \text{KCL}$$

$$\Rightarrow \frac{V_1}{200} - 3\left(\frac{V_1 - V_2}{100}\right) = 0$$

$$\Rightarrow V_1 - 6V_1 + 6V_2 = 0$$

$$\Rightarrow -5V_1 = -6V_2 \Rightarrow V_1 = \frac{6}{5}V_2$$

$$\textcircled{2} \quad \frac{V_2 - V_1}{100} - 6 + \frac{V_2}{400} = 0$$

$$4V_2 - 4V_1 - 2400 + V_2 = 0$$

$$5V_2 - 4V_1 = 2400$$

$$\Rightarrow V_x = V_1 - V_2 = 2400 \text{ V}$$

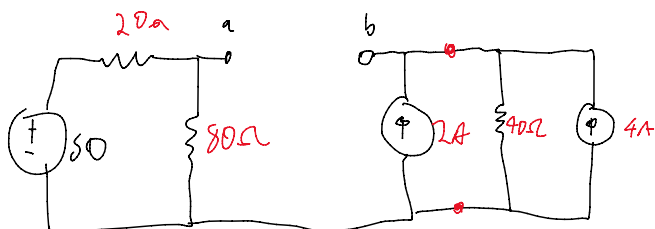
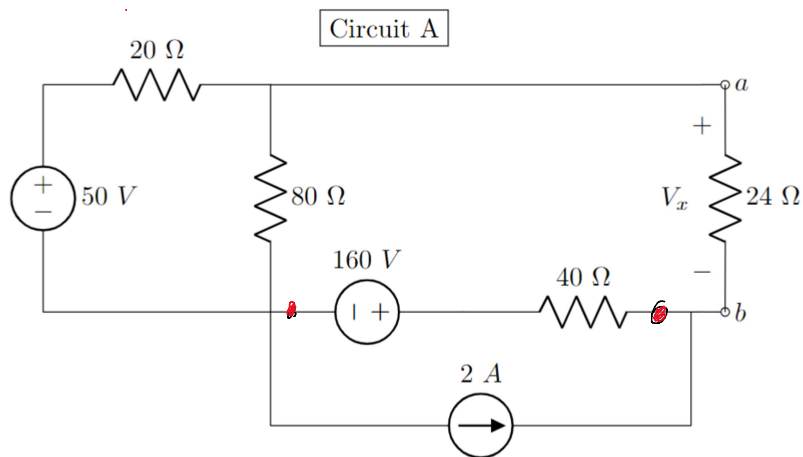
$$\textcircled{1} \text{ into } \textcircled{2} \quad 5V_2 - 4\left(\frac{6}{5}V_2\right) = 2400$$

$$\Rightarrow 0.2V_2 = 2400 \Rightarrow V_2 = 12000$$

$$V_1 = \frac{6}{5}V_2 = 14400$$

Questions 3 to 4 [4]

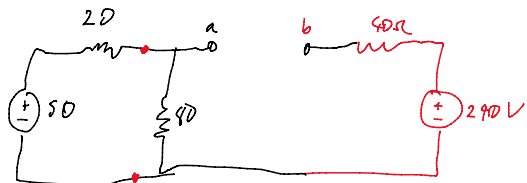
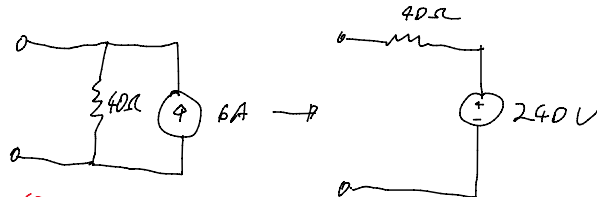
Use source transformation to transform circuit A into circuit B.



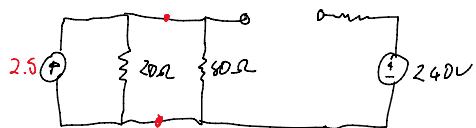
$$v = iR$$

$$i = v/R = 160/40 = 4A$$

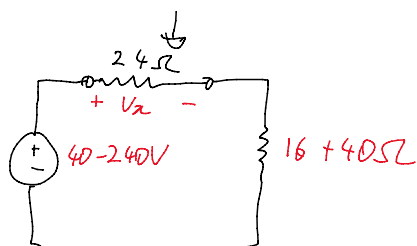
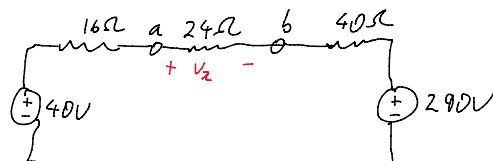
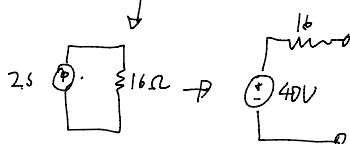
$$i_t = 2A + 4A = 6A$$



$$i = 50/20 = 2.5$$



$$20 \parallel 80 = 16$$

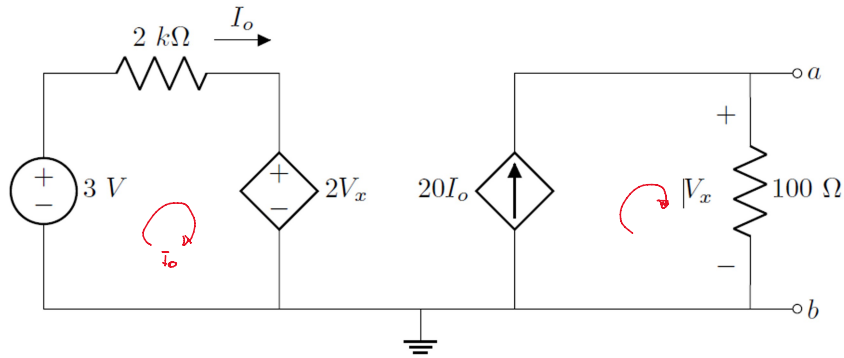


$$B = 40 - 240 = -200V$$

$$R = 16 + 40 = 56\Omega$$

Questions 5 to 6 [4]

Determine the Thevenin equivalent voltage V_{th} and resistance R_{th} with respect to terminals $a - b$.



05 V_{th} [2] 1V

06 R_{th} [2] 33.3

$$V_{t2} = V_x \quad V_{t1} = 20I_o \times 100\Omega$$

$$3 = 2kI_o + 2V_x$$

$$3 = 2kI_o + 2(2000I_o)$$

$$3 = 2kI_o + 4kI_o$$

$$\Rightarrow 3 = -2kI_o$$

$$I_o = \frac{3}{6k} = 0.5mA$$

$$V_x = V_{th} = 20 \times 0.5mA \times 100\Omega = 1V$$

$$V_x = 1V$$

use test source @ 1V $V_x = 1V$

$$3 = 2kI_o + 2V_x$$

$$1 = 2kI_o$$

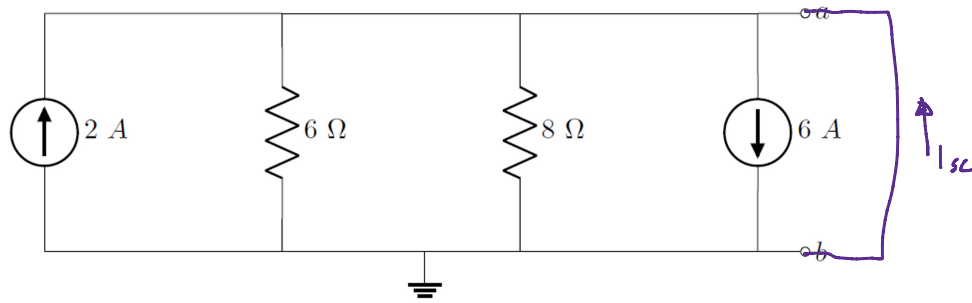
$$I_o = \frac{1}{2k} = 0.5mA$$

$$I_s = 20I_o + \frac{1}{50} = 20(0.5mA) + \frac{1}{50} \\ = 10mA + 20mA \\ = 30mA$$

$$R_{th} = \frac{V}{I} = \frac{1}{30mA} = 33.3\Omega$$

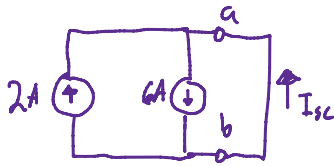
Question 7 [2]

Determine the Norton equivalent current I_N with respect to terminals $a - b$.



07 I_N [2] $-4A$

I_N : short-circuit current across terminals $a-b$



$$I_N = I_{sc} = 2 - 6 = -4A$$

Question 8 [2]

A solar panel has an open circuit voltage of 24 volts and a short circuit current of 2 amperes. What is the maximum power P_{max} that can be transferred by the solar panel?

08 P_{max} [2]

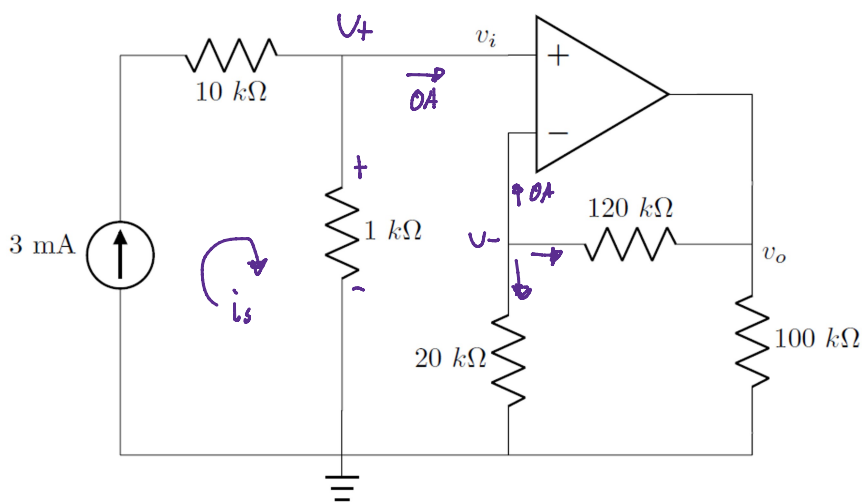
$$R = \frac{V_{oc}}{I_{sc}} = \frac{24}{2} = 12\Omega$$

Max power when $\rightarrow R_L = R \therefore R_L = 12\Omega$

$$P_{max} = \frac{V^2}{4R_L} = \frac{24^2}{4 \times 12} = \frac{576}{48} = 12W$$

Question 9 [2]

Assume the operational amplifier is ideal. Calculate v_o/v_i for the operational amplifier.



09

v_o/v_i [2]

7 (V/V)

$$v_+ = 1k \times 3mA = 3V$$

$$v_+ = v_-$$

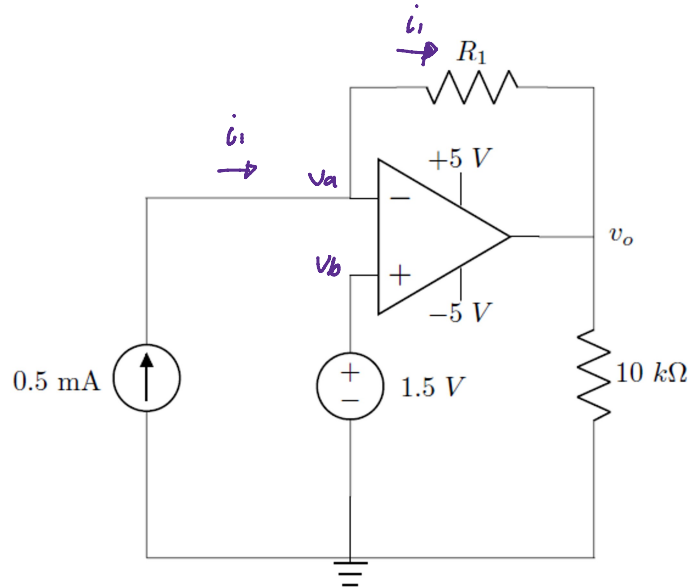
$$\frac{v_- - v_o}{120k} + \frac{v_-}{20k} = 0 \rightarrow \frac{v_- - v_o}{120k} = -\frac{v_-}{20k} \rightarrow v_- - v_o = -\frac{120k}{20k} v_-$$

$$\rightarrow -v_o = -v_- - \frac{120k}{20k} v_- \rightarrow v_o = \left(1 + \frac{120}{20}\right) v_- \quad \therefore v_o = (1 + 6)(3V) = 7 \times 3 = \underline{21V}$$

$$v_i = v_+ = 3V \quad \therefore \frac{v_o}{v_i} = A = \underline{7 \text{ V/V}}$$

Question 10 [2]

Assume the operational amplifier is ideal. Determine R_1 just before the operational amplifier goes into negative saturation.



10

R_1 [2]

$7k\Omega$

KCL at v_a

$$0 = i_1 + \frac{v_a - v_o}{R_1} \rightarrow -i_1 R_1 = v_a - v_o \rightarrow v_o = v_a + i_1 R_1$$

$v_a = v_b = 1.5V$
 $i_1 = 0.5mA$
 $\rightarrow v_o = 1.5 + (0.5mA) R_1$

at positive saturation $v_o = +5V$

1/2 marks

$$5 = 1.5 + (0.5mA) R_1 \rightarrow 3.5 = +0.5mA R_1 \rightarrow R_1 = \frac{3.5}{0.5mA} = 7k\Omega$$

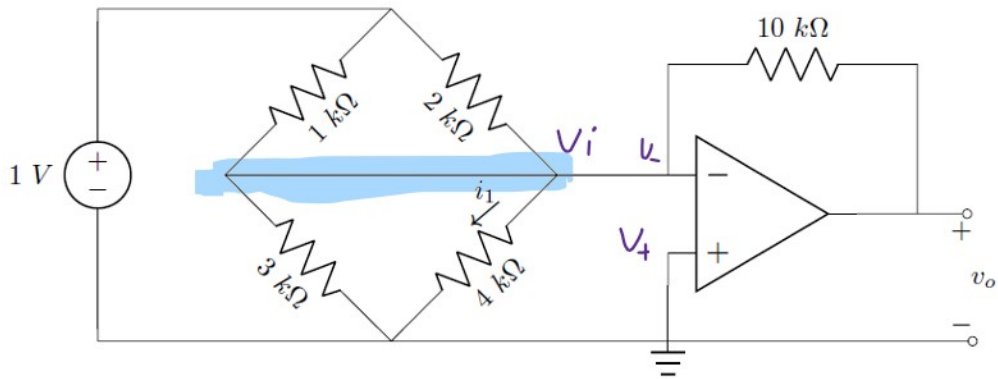
* at negative saturation $v_o = -5V$

$$-5 = 1.5 + (0.5mA) R_1 \rightarrow -7.5 = +0.5mA R_1 \rightarrow R_1 = \frac{-7.5}{0.5mA} = -13k\Omega$$

$-13k\Omega$ means current in other direction $\therefore R_1 = 13k\Omega$

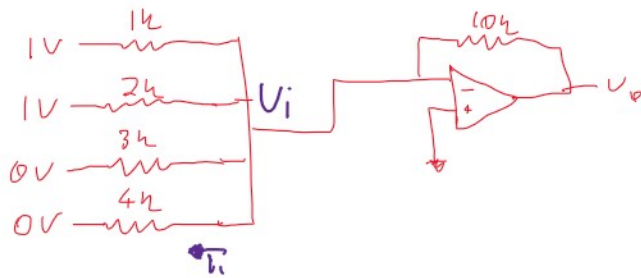
Questions 11 to 12 [4]

Assume that the operational amplifier is ideal. Determine i_1 and v_o .



11 i_1 [2] 0A

12 v_o [2] -15V



$$v_+ = v_- = v_i = 0V$$

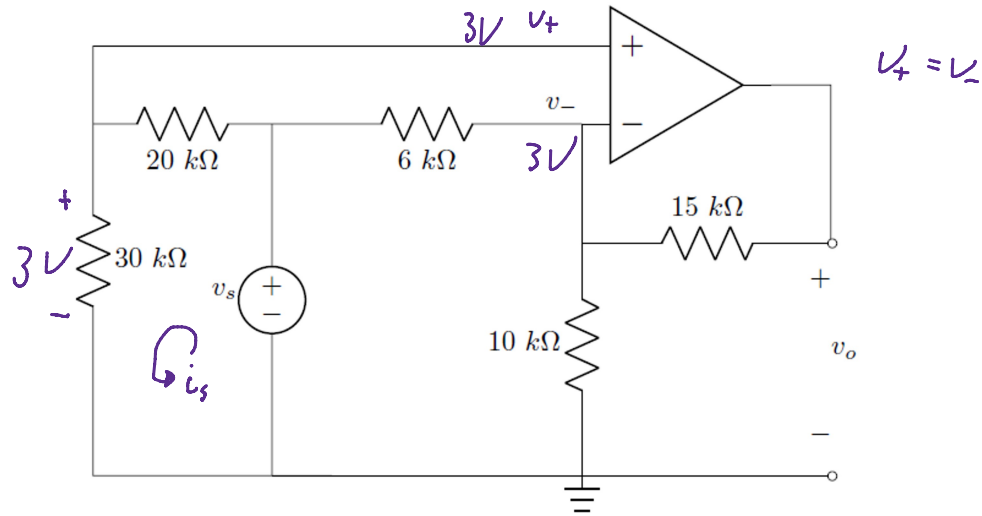
$$\therefore i_1 = \frac{v_i - 0}{4k} = \underline{0A}$$

$$v_o = - \left(\frac{10k}{1k} (1) + \frac{10k}{2k} (1) \right) = -(10 + 5) = \underline{-15V}$$

Questions 13 to 14 [4]

Assume that the operational amplifier is ideal.

Given $v_- = 3\text{ V}$, determine v_s and v_o .



13 v_s [2] 5V

14 v_o [2] 2.5V

13

$$3V = \frac{30k}{30k + 20k} \times v_s$$

$$v_s = \frac{30k + 20k}{30k} \times 3$$

$$v_s = \frac{50k}{30k} \times 3$$

$$v_s = 5V$$

KCL @ v_b

$$\frac{v_b - 0}{10k} + \frac{v_b - v_o}{15k} + \frac{v_b - v_s}{6k} = 0$$

$$3v_b + 2v_b - 2v_o + 5v_b - 5v_s = 0$$

$$-2v_o = -10v_b + 5v_s$$

$$v_o = 5v_b - 2.5v_s$$

$$v_o = 5(3) - 2.5(5)$$

$$= 15 - 12.5 = \underline{2.5V}$$

14

KCL @ v_b

14

$$\frac{v_b - 0}{10k} + \frac{v_b - v_o}{15k} + \frac{v_b - v_s}{6k} = 0 \quad \times 15k$$

$$\frac{v_b}{R_L} + \frac{v_b - v_o}{R_f} + \frac{v_b - v_s}{R_H} = 0$$

$$\frac{R_f}{R_L} v_b + v_b - v_o + \frac{R_f}{R_H} v_b - \frac{R_f}{R_H} v_s = 0$$

$$-v_o = -v_b \left(1 + \frac{R_f}{R_L} + \frac{R_f}{R_H} \right) + \frac{R_f}{R_H} v_s$$

$$\Rightarrow v_o = \left(1 + \frac{R_f}{R_L} + \frac{R_f}{R_H} \right) v_b - \frac{R_f}{R_H} v_s$$

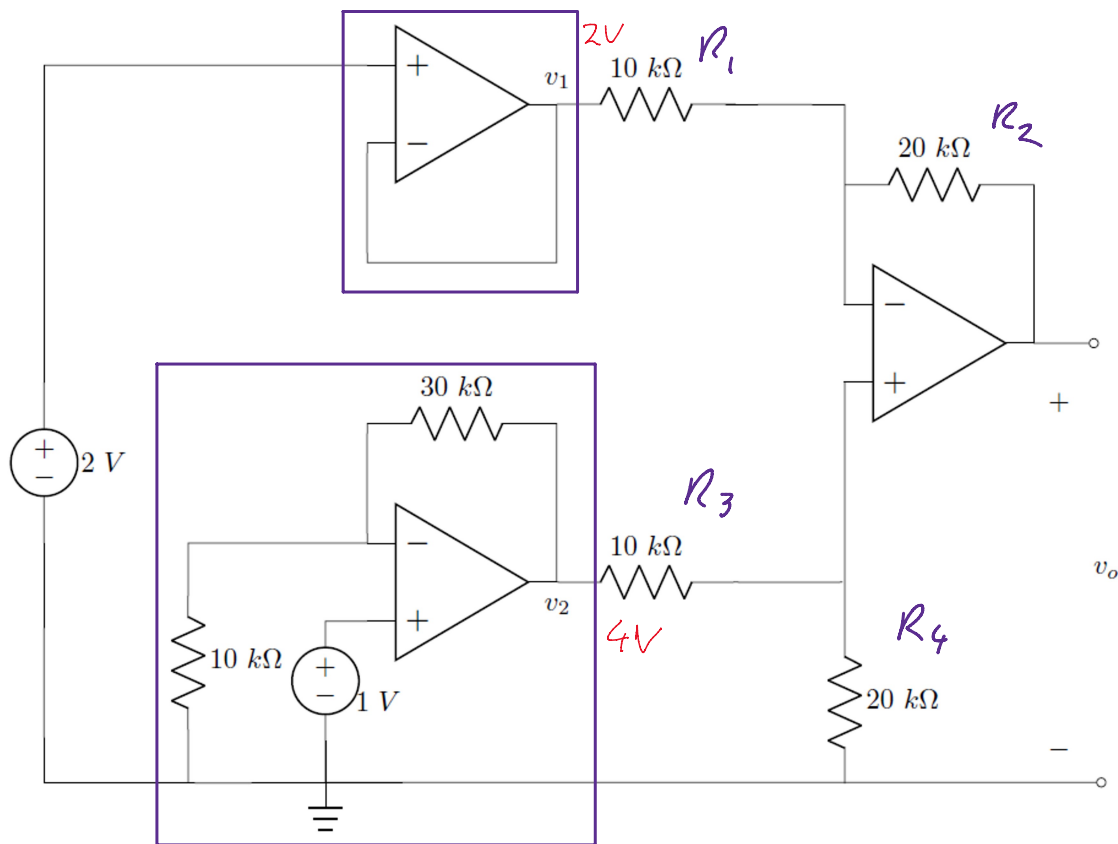
$$= \left(1 + \frac{15k}{10k} + \frac{15k}{6k} \right) (3) - \frac{15k}{6k} (5)$$

$$= (1 + 1.5 + 2.5)(3) - 2.5(5)$$

$$= 5 \times 3 - 2.5 \times 5 = 15 - 12.5 = \underline{2.5V}$$

Question 15 [2]

Assume that the operational amplifier is ideal. Determine v_o .



15 v_o [2] 4V

$$v_1 = 1 \times 2V = 2V$$

v_2 is a non-inverting op-amp

$$v_2 = \left(1 + \frac{R_f}{R_i}\right) v_i \text{ with } R_f = 30k, R_i = 10k, v_i = 1V$$

$$v_2 = \left(1 + \frac{30k}{10k}\right) \times 1V = (1 + 3) \times 1V = 4 \times 1V = \underline{4V}$$

$$\underline{v_2 = 4V}$$

v_o is a difference op-amp

$$\therefore v_o = \frac{R_2}{R_1} \frac{(1 + R_1/R_2)}{(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

$$R_1 = R_3 = 10k\Omega, \quad R_2 = R_4 = 20k\Omega$$

$$\text{but } \frac{R_1}{R_2} = \frac{R_3}{R_4} \therefore v_o = \frac{R_2}{R_1} (v_2 - v_1)$$

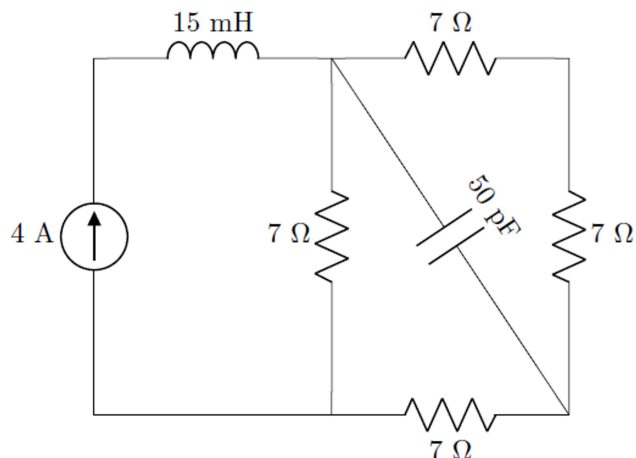
$$\rightarrow v_o = \frac{20k}{10k} (v_2 - v_1) = 2(4 - 2)$$

$$v_o = 4V$$

Section 3

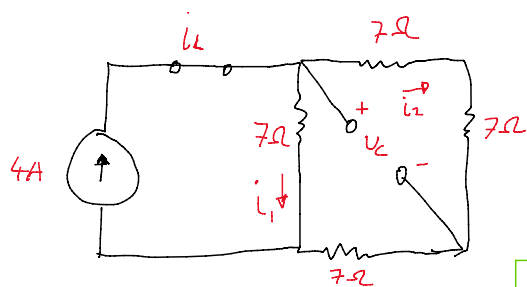
Questions 16 to 17 [4]

Under DC conditions, calculate w_C , the energy stored in the capacitor and w_L , the energy stored in the inductor.



16 w_C [2]

17 w_L [2]



$$i_L = 4A \quad w_L = \frac{1}{2} L i_L^2 = \frac{1}{2} (15m) 4^2$$

$$= \frac{4^2}{2} \times 15m$$

$$w_L = 120mJ$$

$$v_C = i_2 (7+7)$$

$$i_L = i_1 + i_2$$

$$i_1 = \frac{7\Omega}{7\Omega + (7+7+7)} \times i_L$$

$$= \frac{7}{28} \times 4 = 1A$$

$$\therefore v_C = 1 \times 14 = 14V$$

$$v_C = 14V \quad w_C = \frac{1}{2} C v_C^2 = \frac{1}{2} (50p)(14)^2$$

$$= \frac{14^2}{2} (50p)$$

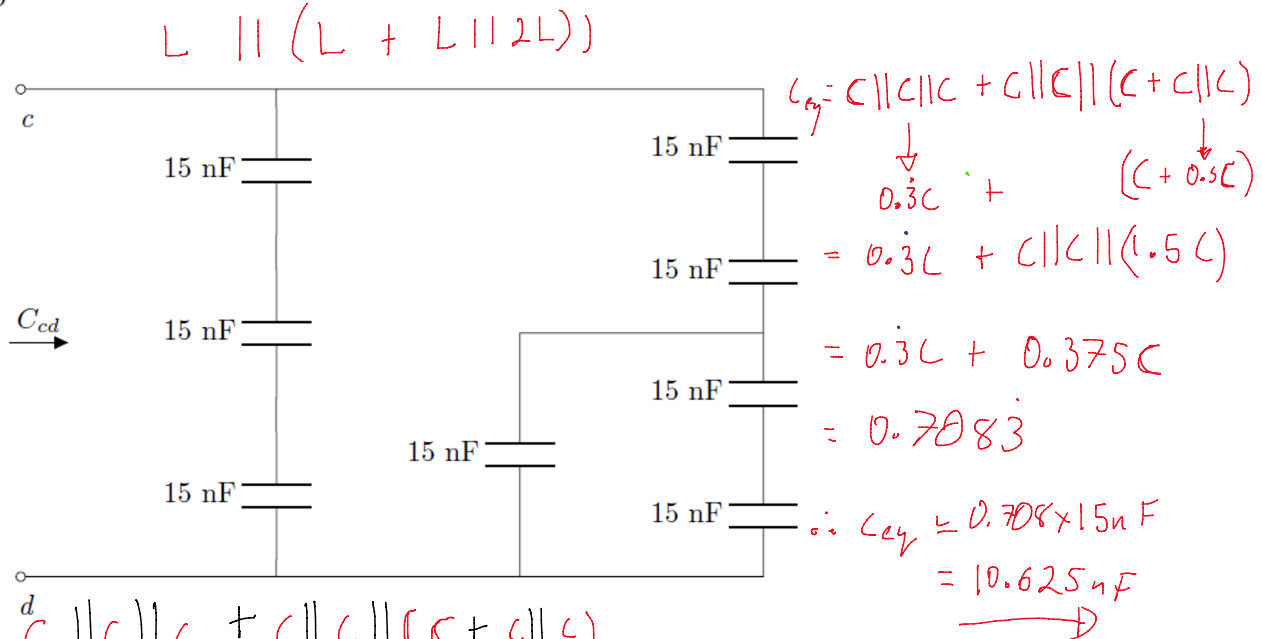
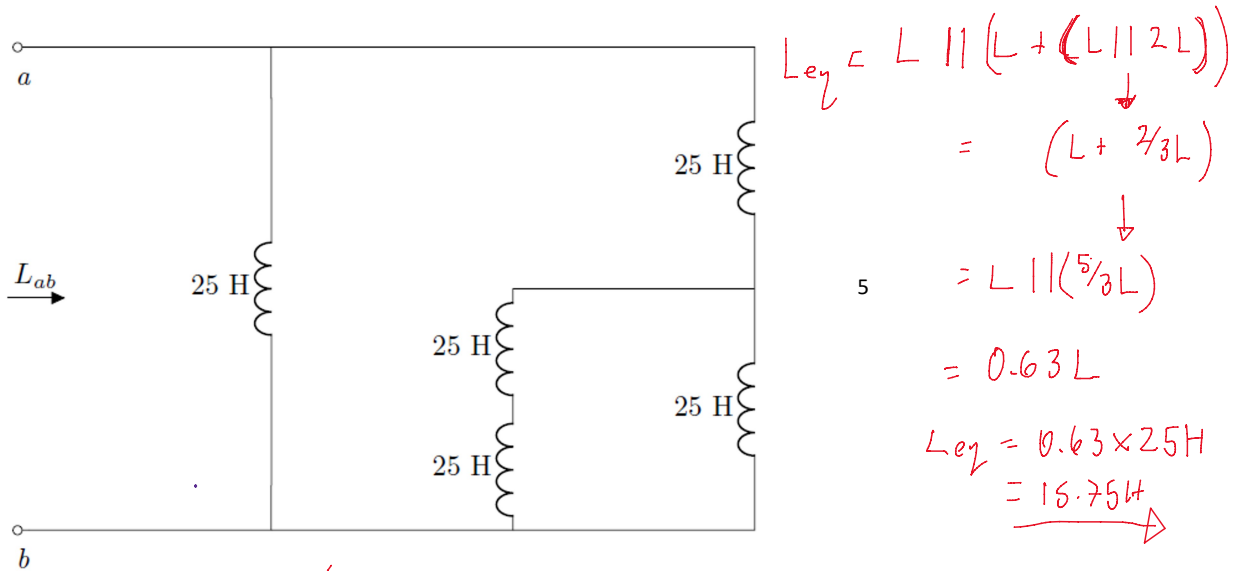
$$= 98 (50p)$$

$$= 0.0049J$$

$$w_C = 4.9nJ$$

Questions 18 to 19 [4]

What is the equivalent inductance, L_{ab} , and the equivalent capacitance, C_{cd} , for the circuits below?



$$C \parallel C \parallel C + C \parallel C \parallel (C + C \parallel C)$$

$$5 + 5.625 = 10.625nF$$

18	L_{ab} [2]	15.75H
19	C_{cd} [2]	10.625nF

Questions 20 to 22 [4]

Three sinusoids are given as follows:

$$V_1 = 7 \angle 60^\circ - 90^\circ = 7 \angle -30^\circ$$

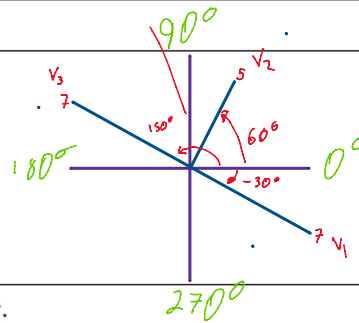
$$V_2 = 5 \angle 60^\circ$$

$$V_3 = -7 \angle -30^\circ = 7 \angle 150^\circ$$

$$v_1(t) = 7 \sin(200t + 60^\circ) \text{ V}$$

$$v_2(t) = 5 \cos(200t + 60^\circ) \text{ V}$$

$$v_3(t) = -7 \sin(200t + 60^\circ) \text{ V}$$



20 Consider $v_1(t)$ and $v_2(t)$ and determine which lags the other.

X

- Answer = 1 if $v_1(t)$ lags $v_2(t)$.

- Answer = 2 if $v_2(t)$ lags $v_1(t)$.

Write 1 or 2 in the answer block. [1]

1

21 Consider $v_2(t)$ and $v_3(t)$ and determine which lags the other.

- Answer = 1 if $v_2(t)$ lags $v_3(t)$.

- Answer = 2 if $v_3(t)$ lags $v_2(t)$.

Write 1 or 2 in the answer block. [1]

1

22 The sinusoids can be added as follows:

- $v_0(t) = v_1(t) + v_2(t) + v_3(t)$

$$V_0 = 7 \angle -30^\circ + 5 \angle 60^\circ - 7 \angle -30^\circ = 5 \angle 60^\circ$$

Determine the phasor representation V_0 of $v_0(t)$. [2]

Question 23 [2]

Simplify the following expression for Z .

$$Z = \frac{j15}{(15 + j6)\sqrt{(6 + j1)}}$$

23 Z [2]

$$0.1681678 + 0.3368128i$$

$$-0.1681678 - 0.3368128i$$

$$= 0.376 \angle 63.28^\circ$$

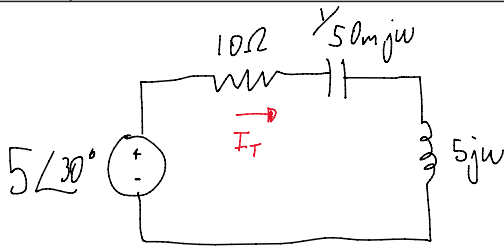
$$\frac{15 \angle 90^\circ}{16.16 \angle 21.8^\circ \times \sqrt{6.08 \angle 9.46^\circ}} = \frac{15 \angle 90^\circ}{16.16 \angle 21.8^\circ \times 2.47 \angle \pm 4.56^\circ} = \frac{15}{16.16 \times 2.47} \angle 90 - (21.8 \pm 4.56)$$

$$\rightarrow 0.376 \angle 63.64^\circ \text{ or } 0.376 \angle 72.76^\circ$$

Question 24 [2]

4.33 + 2.50j

A voltage source $v_s(t) = 5 \cos(4t + 30^\circ) \text{ V}$ is connected in series with a 10Ω resistor, 50 mF capacitor, and 5 H inductor. The total current, $i_T(t)$, in the circuit is measured in the direction as it exits the positive terminal of the voltage source. Determine the phasor representation $\mathbf{I}_T$ of the current $i_T(t)$.

24 $\mathbf{I}_T$ [2]

$$V = IZ \therefore I = V/Z$$

$$\omega = 4$$

$$I = \frac{5\angle 30^\circ}{10 + \frac{1}{50\text{m} \times 4j} + 5(4j)} = \frac{5\angle 30^\circ}{10 - 5j + 20j}$$

$$= \frac{5\angle 30^\circ}{10 + j15} = \frac{5\angle 30^\circ}{18 \angle 56.31^\circ} = 0.27 \angle -26.18^\circ$$

Answer

Unit

SECTION 1		
01	$V_x^V =$	$-30 \quad V$
02	$V_x^I =$	$2400 \quad \cancel{+60} \quad V$
03	$B =$	$-200 \quad \text{No unit required}$
04	$R_B =$	$56 \quad \Omega$
05	$V_{th} =$	$1 \quad V$
06	$R_{th} =$	$33.3 \quad \Omega$
07	$I_N =$	$-4 \quad A$
08	$P_{max} =$	$12 \quad W$
SECTION 2		
09	$v_o/v_i =$	$7 \quad \text{No unit required}$
10	$R_1 =$	$13k \quad \Omega$
11	$i_1 =$	$0 \quad A$
12	$v_0 =$	$-15 \quad V$
13	$v_s =$	$5 \quad V$
14	$v_0 =$	$2.5 \quad V$
15	$v_0 =$	$4 \quad V$
SECTION 3		
16	$\omega_C =$	$120m \quad J$
17	$\omega_L =$	$4.9n \quad J$
18	$L_{ab} =$	$15.75 \quad H$
19	$C_{cd} =$	$10.625 \quad nF$
20	1 or 2	$1 \quad 2 \quad \text{No unit required}$
21	1 or 2	$1 \quad \text{No unit required}$
22	$V_0 =$	$5 \angle 60 \quad V$
23	$Z =$	$\underline{0.376 \angle 63.64^\circ} \text{ or } 0.376 \angle 72.76^\circ \quad \cancel{0.376 \angle 63.28^\circ} \quad \text{No unit required}$
24	$I_T =$	$0.27 \angle -26.18^\circ \quad \cancel{0.0399 \angle -55.25^\circ} \quad A$